

Synopsis

End-to-End Machine Learning Regression System for Institutional Energy Usage Prediction

Problem Id: 23

Name : Vinamra Semwal

Class Roll No. :76

University Roll No: 2028907

Section – K G2

Name : Mayank Fikwal

Class Roll No. :53

University Roll No: 2028764

Section – K G2

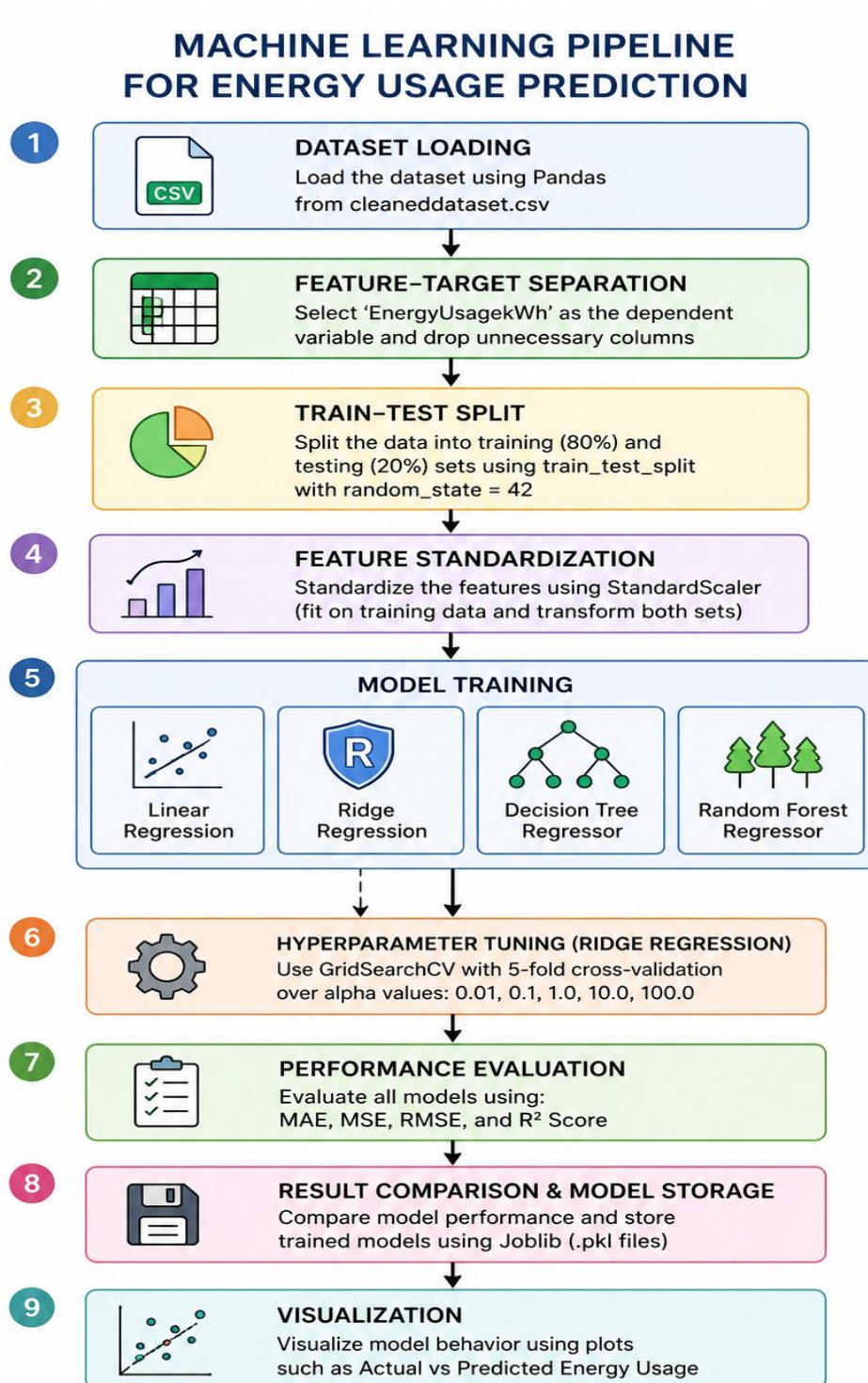
1. Problem Description.....	3
2. Architecture	3
3. Models model.....	4
4. Evaluation metrics used.....	5
5. Preliminary Results and Visualizations	5
6. References.....	6

1. Problem Description

The Student Energy Usage Prediction System is a Python-based machine learning project developed to predict energy consumption values from a cleaned dataset using supervised learning models. The implementation in the training notebook loads `cleaneddataset.csv`, removes non-predictive columns such as `Timestamp`, `Date`, `EnergyUsagekWh`, and `Month`, and uses the remaining variables as input features while `EnergyUsagekWh` is treated as the target variable. The project applies preprocessing, train-test splitting, feature scaling, model training, evaluation, comparison, and visualization in order to support accurate data-driven prediction.

This work reduces manual analysis by automatically learning patterns from historical data and estimating future energy usage values. Such a system can help in understanding usage behavior, improving planning, and supporting efficient monitoring of energy-related trends through machine learning techniques.

2. Architecture



3. Models Used

The notebook contains four regression models for prediction

Linear Regression

Linear Regression is used as the baseline model to capture a linear relationship between the input variables and energy usage. The trained model achieved MAE of 2.459351, RMSE of 3.060184, and R2 score of 0.995317 on the test set.

Ridge Regression

Ridge Regression is a regularized linear model that helps reduce overfitting by penalizing large coefficients. Grid search selected the best alpha value as 0.01, and the tuned model achieved MAE of 2.459351, RMSE of 3.060183, and R2 score of 0.995318, which was the best overall performance among the tested models.

Decision Tree Regressor

Decision Tree Regressor is a non-linear model that learns prediction rules through recursive splitting of the dataset. In the notebook, this model produced MAE of 4.231730, RMSE of 5.587901, and R2 score of 0.984387 on the test data.

Random Forest Regressor

Random Forest Regressor is an ensemble model that combines predictions from multiple decision trees to improve robustness and generalization. With `n_estimators=100` and `random_state=42`, the model obtained MAE of 2.978189, RMSE of 3.808416, and R2 score of 0.992748.

4. Evaluation Metrics Used

The notebook uses four evaluation metrics to judge regression performance.

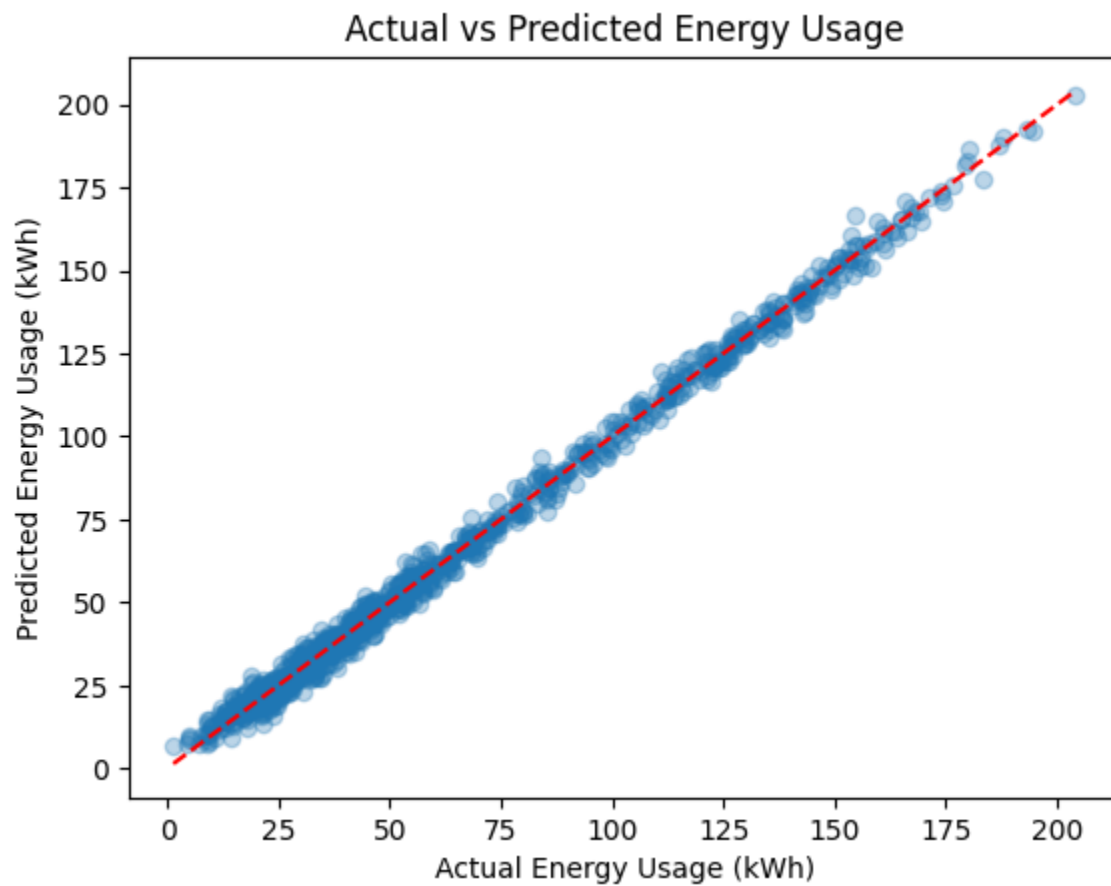
- **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted values.
- **Mean Squared Error (MSE):** Measures the average squared prediction error and gives more weight to large errors.
- **Root Mean Squared Error (RMSE):** Represents the square root of MSE and expresses error in the same unit as the target variable.
- **R2 Score:** Indicates how well the model explains the variance in the target variable; a value closer to 1 shows better fit.

5. Preliminary Results and Visualizations

The notebook compares all implemented models and shows that Ridge Regression and Linear Regression provide the strongest predictive performance on the dataset. Ridge Regression gave the highest R2 score of 0.995318, while Linear Regression produced nearly identical results with R2 score of 0.995317. Random Forest also performed well with R2 score of 0.992748, whereas Decision Tree showed comparatively lower accuracy with R2 score of 0.984387.

Model	MAE	RMSE	R2 Score	Saved Model File
Linear Regression	2.459351	3.060184	0.995317	linearregressionmodel.pkl
Ridge Regression	2.459351	3.060183	0.995318	ridgeregressionmodel.pkl
Decision Tree	4.231730	5.587901	0.984387	decisiontreemodel.pkl
Random Forest	2.978189	3.808416	0.992748	randomforestmodel.pkl

The visualization in the notebook includes a scatter plot of actual versus predicted energy usage values, which helps verify how closely predictions align with true observations. A reference diagonal line is also plotted to show ideal prediction behavior, and the closeness of points to that line indicates high prediction quality.



6. References

- Phase 2 base report structure and topic outline from the attached report file.
- Python libraries used in implementation: Pandas, NumPy, Matplotlib, scikit-learn, and Joblib.
- Models and outputs documented in the training notebook, including Linear Regression, Ridge Regression, Decision Tree Regressor, and Random Forest Regressor.